



COMP4010: Data Visualization

ToxicDDI Explorer: Visualizing Toxic Drug-Drug Interactions by Type and Severity

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TABLE OF CONTENTS

PROJECT OVERVIEW

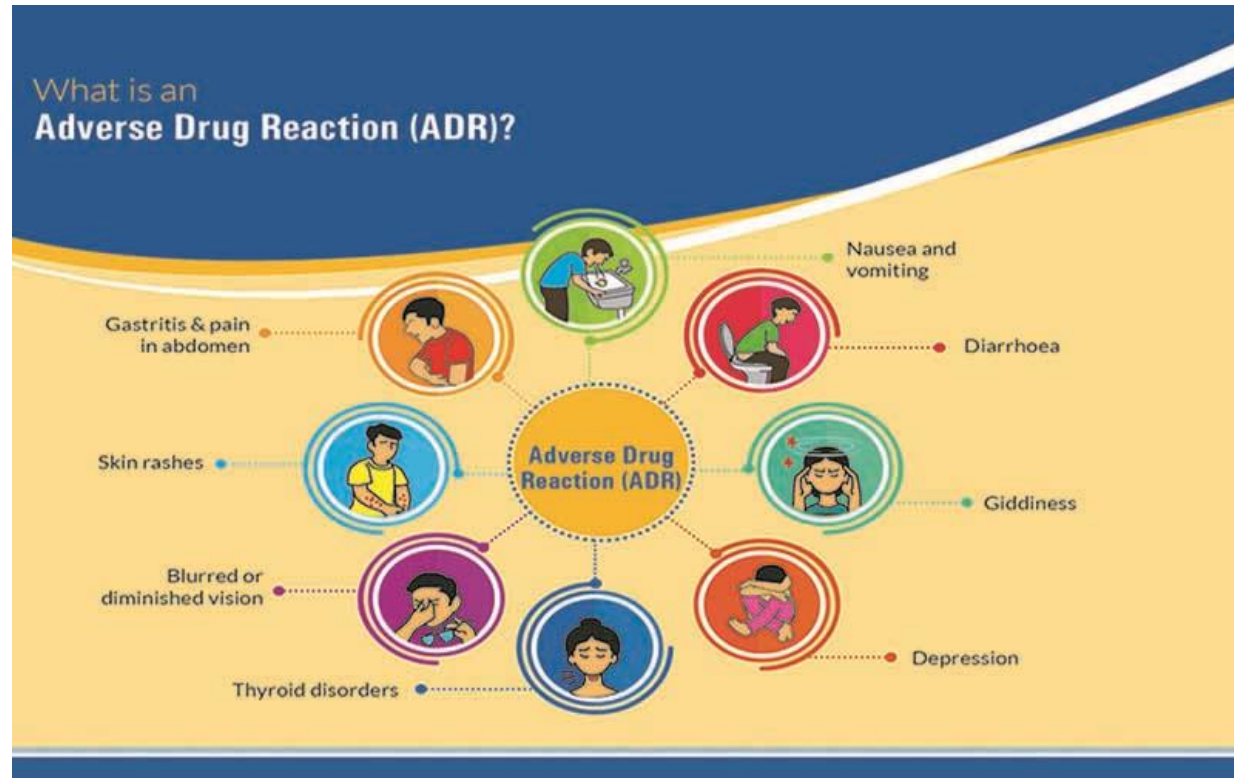
DATA PREPROCESSING

DATA VISUALIZATION & DASHBOARD DEMO

INSIGHTS & DISCUSSION

LIMITATIONS & FUTURE DIRECTIONS

Motivation



PROJECT OVERVIEW

❖ Questions:

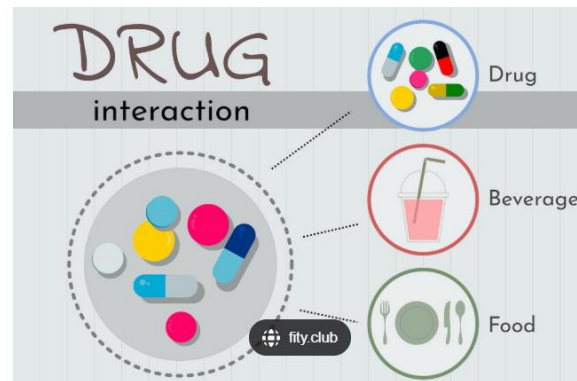
How can we effectively visualize toxic drug-drug interactions (DDIs) to support safer prescribing practices and improve understanding of toxicity risks across different drug combinations?

❖ Project Objectives:

- Develop an interactive dashboard that visualizes toxic DDIs.
- Leverage a dataset derived from trusted sources
- Provide detailed breakdowns of individual drugs' interaction profiles.

❖ Key functionalities

- **Interactive Network Graphs:** Visualizing DDIs as interconnected nodes and edges, with filters for toxicity type and severity.
- **Multi-View Exploration:** The application includes three distinct views:
 - (1) a network-wide overview of all interactions, (2) a focused view of interactions for a single drug, and (3) a customizable drug pair query tool.
- **Dynamic Filtering and Customization:** Users can filter by severity, toxicity type, or drug of interest; view interaction descriptions; and search interactively—enhancing usability and insight generation.



DATA PREPARATION & PROCESSING

❖ Data Collection & Normalization

- **Data sources**
 - **Mendeley Dataset** (Yu et al., 2020): extracted from DrugBank data for toxic DDI, contains Raw DDI records and Toxicity mapping table
 - **DDInter Database** (Tian et al., 2025): contains Clinical severity levels, Descriptions of interactions, Management recommendations
- **Data Preparation steps:**



- **Outputs:** ``ddi_with_ddinter_info.csv``
contains **2,144 toxic DDI entries** with **10 attributes**: Drug Identifiers and Names, Interaction types, matched toxicity categories, and enriched clinical annotations from the DDInter database.

DATA PREPARATION & PROCESSING

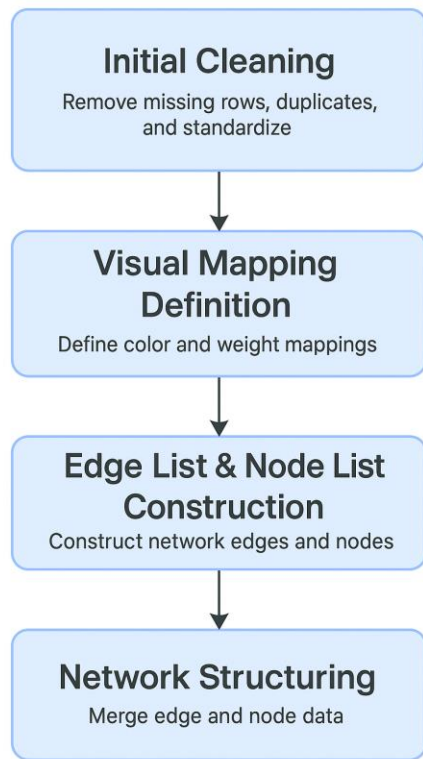
❖ Data Processing & Network structuring

○ Processing Steps:

```
# === Color for interaction types ===
interaction_color_map <- list(
  cardiotoxicity      = "#FF4C4C",
  nephrotoxicity      = "#00CFFF",
  hepatotoxicity      = "#FFA947",
  neurotoxicity       = "#B266FF",
  ototoxicity         = "#FFD700",
  pulmonary_toxicity  = "#3CFFB3",
  unknown             = "#666666"
)
```

○ Outputs: `network\_data.rds`

The complete processed network containing all necessary metadata, including edges, nodes, and styling information, for integration into the Shiny dashboard.



DATA VISUALIZATION APPROACH

❖ Design Rationales

○ Modular View Design

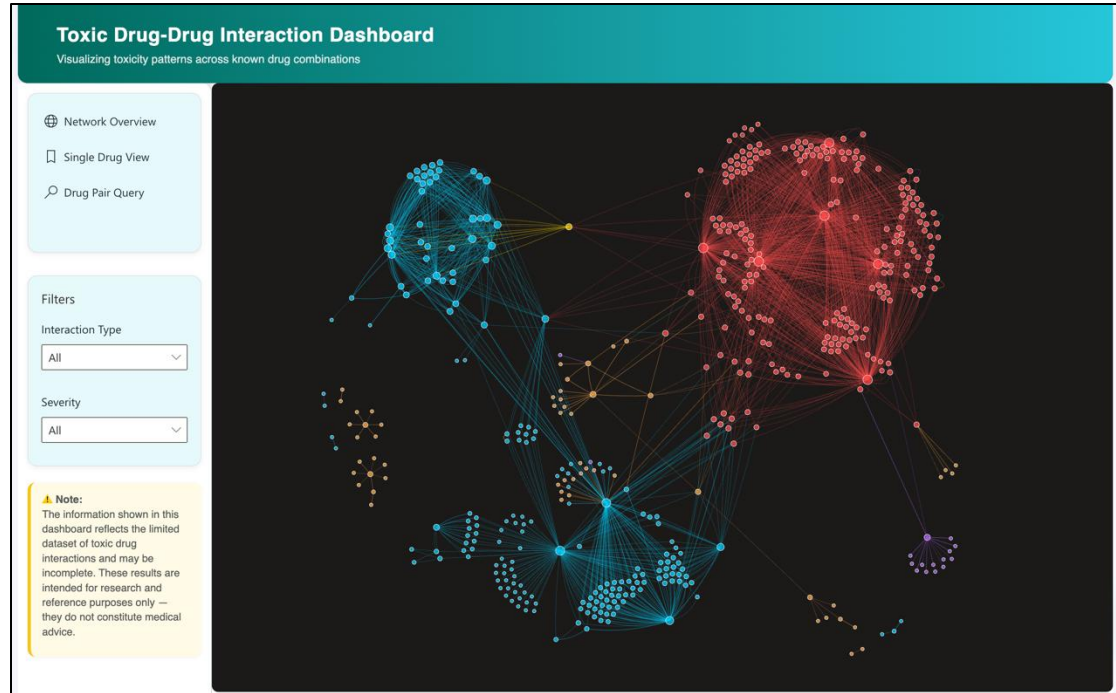
- 3 semantically linked views for global, focused, and specific queries.
- Supports exploratory and task-oriented workflows.

○ Guiding Principles:

- **Grammar of Graphics:** Consistent visual encoding.
- **Shneiderman's Mantra:** Overview first → zoom/filter → details on demand.
- **Perceptual Effectiveness:** Thickness for severity, hue for type, size for importance.

○ Purpose-driven Views:

- **Network Overview:** Highlights global structure and toxic clusters.
- **Single Drug View:** Shows detailed drug-centered interactions and risk breakdowns.
- **Drug Pair Query:** Offers clinical lookup in card format with severity tagging.



DATA VISUALIZATION APPROACH

❖ Visualization Techniques and Visual Encoding

o **Network Overview:**

- o Diagram type: Force-directed graph
- o Encoding:
 - o Edge thickness → severity
 - o Edge color → toxicity type
 - o Node size → degree centrality
 - o Node color → dominant toxicity

o **Single Drug View:**

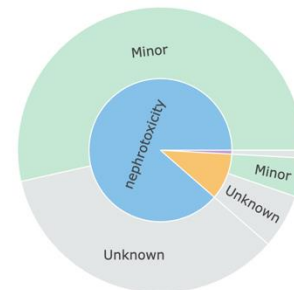
- o Local interaction subgraph centered on selected drug.
- o Sunburst chart:
 - o Inner ring: toxicity type
 - o Outer ring: severity distribution
 - o Hierarchical part-to-whole breakdown

o **Drug Pair Query:**

- o Card layout for selected drug pairs
- o Elements: Drug IDs, toxicity, severity, Clinical interaction text, Severity tag with color emphasis



Interaction Severity Distribution



DATA VISUALIZATION APPROACH

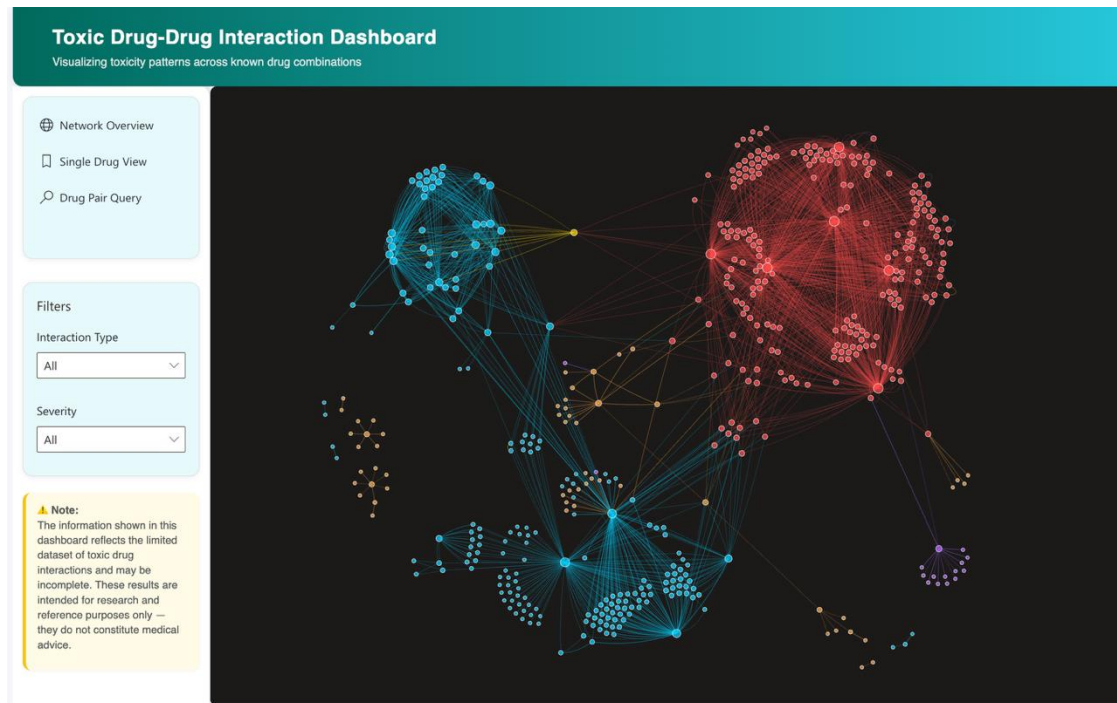
❖ Libraries and tools used

Library/Tool	Purpose
<code>visNetwork</code>	Main engine for network graphs – interactive layout, styles, tooltips
<code>plotly</code>	Sunburst chart for severity distribution – interactivity & hierarchy
<code>dplyr</code> , <code>tidyr</code>	Data wrangling: filtering, grouping, joining, reshaping
<code>scales</code>	Visual normalization: node size, edge weight (<code>rescale()</code> function)
<code>shiny.fluent</code> , <code>react</code>	Fluent Design UI: dropdowns, filters, cards, layout via <code>Stack</code> , <code>Text</code> , etc.

DASHBOARD DEMO

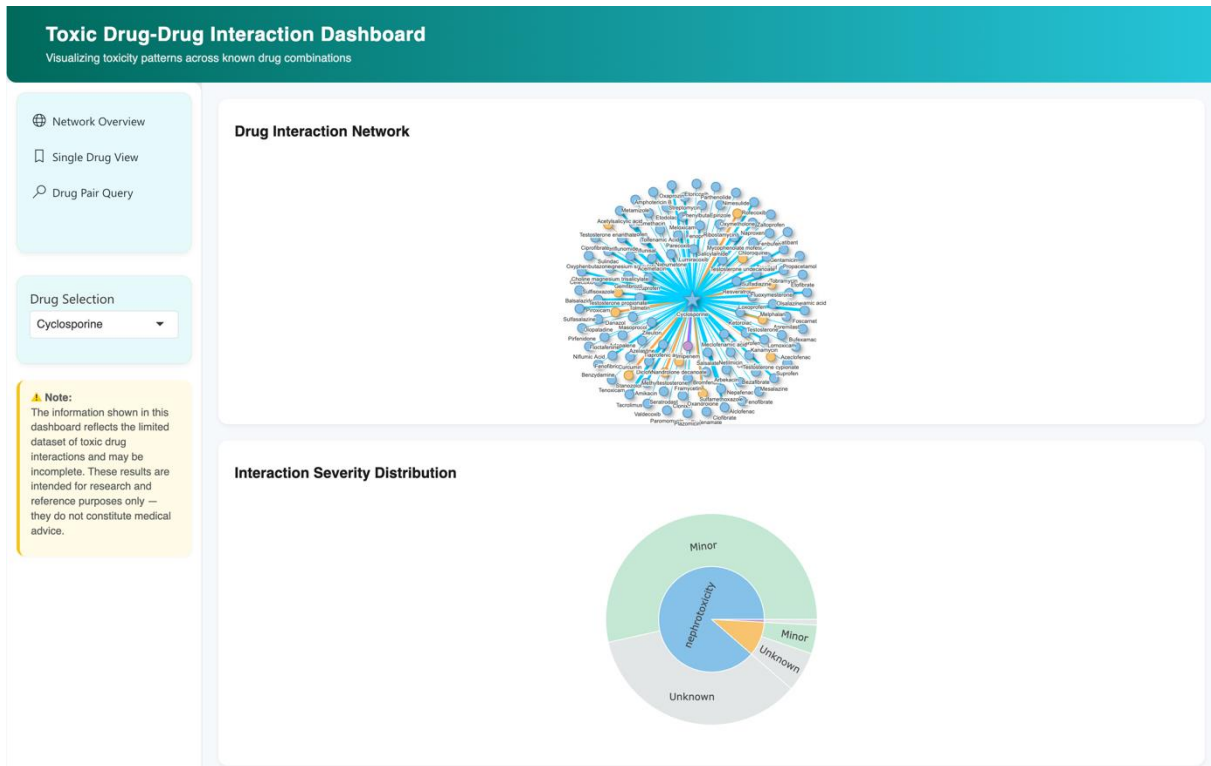
FINAL PRODUCT

❖ Network Overview



FINAL PRODUCT

❖ Single Drug View



FINAL PRODUCT

❖ Drug Pair Query

Toxic Drug-Drug Interaction Dashboard

Visualizing toxicity patterns across known drug combinations

[Network Overview](#)
[Single Drug View](#)
[Drug Pair Query](#)

Note:
The information shown in this dashboard reflects the limited dataset of toxic drug interactions and may be incomplete. These results are intended for research and reference purposes only – they do not constitute medical advice.

Select 2–5 Drugs:

Cyclosporine Oxymetholone Zileuton

Interaction 1

Drug 1: DB000091 (Cyclosporine)
Drug 2: DB06412 (Oxymetholone)
Type: Hepatotoxicity
Severity: Moderate
Description: Coadministration with androgens or anabolic steroids may increase the blood concentrations of cyclosporine. The exact mechanism of interaction is unknown but may involve competitive inhibition of the CYP450 3A4 metabolism of cyclosporine.

Interaction 2

Drug 1: DB000091 (Cyclosporine)
Drug 2: DB00744 (Zileuton)
Type: Nephrotoxicity
Severity: Minor
Description: Zileuton is believed to be a weak inhibitor of CYP450 3A4 enzymes, of which cyclosporine is a known substrate. Although no studies have been conducted to verify or quantify the interaction, caution is recommended when using this combination until further data is available.

INSIGHTS & DISCUSSION

❖ Data Insights from the Visualization of Toxic Drug-Drug Interactions

- **Data Overview:**
 - Dataset: 503 unique drugs, 2,143 toxic drug-drug interactions (DDIs).
 - Clinically meaningful network for insights
- **Toxicity Type Distribution (Highly Imbalanced):**
 - **Major Types (95%+):** Cardiotoxicity (60%), Nephrotoxicity (35%).
 - **Less Common:** Hepatotoxicity (67 interactions), Neurotoxicity (19 interactions).
 - **Rare:** Ototoxicity, Pulmonary toxicity (1 recorded case).
 - **Implication:** May reflect true prevalence or data reporting biases; highlights reporting gaps.
 - Dense communities for major types; isolated nodes for less frequent types.
- **"Hub Drugs":**
 - High-degree nodes involved in many toxic interactions.
 - Warrant closer clinical investigation due to potential for compounded toxicity in polypharmacy.
 - Aid in hypothesis generation and clinical prioritization.

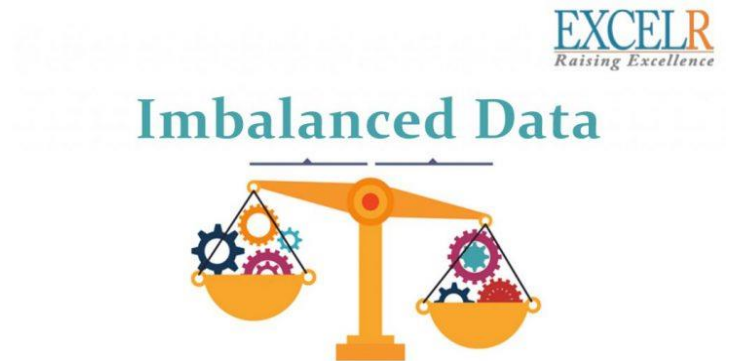
INSIGHTS & DISCUSSION

❖ Effectiveness of the Visualization Techniques

- **Overall:** Optimized for structural exploration, drug profiling, interaction lookup; visual encodings match data type and perceptual effectiveness.
- **Network View:**
 - Toxicity type: Distinct hues (categorical).
 - Severity: Edge width (ordinal).
 - Node size: Degree centrality (highly interactive drugs).
- **Single Drug View (Sunburst Chart):**
 - Compositional snapshot of interaction types and severity.
 - Nests severity within toxicity for part-to-whole reasoning.
 - Reveals dominant toxicological profile of a single drug
- **Drug Pair Query (Card-based Layout):**
 - Prioritizes clarity; structured, readable format for key interaction fields (severity, description, type).
 - Visual enhancements: color-coded severity, fallback labels.

LIMITATIONS

- Dataset incomplete; misses rare or emerging toxic interactions (e.g., pulmonary toxicity).
- Imbalanced toxicity type coverage; dominated by cardiotoxicity & nephrotoxicity.
- Severity labels and descriptions sometimes incomplete or inconsistent.
- Network visualization may become cluttered with larger data or fewer filters.
- Dashboard is educational only—not for clinical decision-making.



Future Directions

- ❖ Expand dataset using pharmacovigilance and clinical records to improve coverage and balance.
- ❖ Add interactive features: cross-view highlighting, filtering, annotation, and saving.
- ❖ Develop a web-based platform and integrate with clinical decision-support tools for real-time use.





THANK YOU!